

MKT-07: Pricing Strategy

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Value-Based Pricing, Competitive Analysis, and Revenue Optimization

Document Version: 1.0 | Date: July 2026 | Classification: Investor Confidential

EXECUTIVE SUMMARY

CSM's pricing strategy is built on value-based pricing — the price of a dielectric protection system is set as a fraction of the economic loss it prevents, not as a markup on manufacturing cost. This approach captures a portion of the \$2-7.8 trillion in averted catastrophe loss as revenue for CSM, while leaving the customer with a clear positive economic return.

The average CSM system is priced at 5-15% of the avoided loss over a 10-year period, producing gross margins of 55-65% at scale while providing customers with ROI periods of 3-8 years (or zero years when financed through insurance premium savings).

This document details pricing by customer segment, competitive positioning, volume discount structures, licensing models, geographic variations, and price elasticity estimates.

SECTION 1: VALUE-BASED PRICING METHODOLOGY

1.1 The Economic Rationale

CSM's pricing philosophy: **A product that prevents a \$10 million loss is worth something less than \$10 million.** The question is how much less. CSM's goal is to capture 5-15% of the value created, consistent with pricing for enterprise software, medical devices, and other high-value-add technology products.

Value-Based Price Formula

$$\text{CSM System Price} = (\text{Avoided Economic Loss} \times \text{Value Capture Rate}) \div (1 + \text{Discount Rate})^t$$

Where: - Avoided Economic Loss = cost of damage + business interruption + insurance premium increase + regulatory penalty - Value Capture Rate = CSM's target share of value (5-15%) - Discount Rate = customer's cost of capital (5-12%) - t = time until first event with protection (probability-weighted)

Example: Utility Substation (\$75M replacement value, 2% annual EMP/CME loss probability)

Parameter	Value
Substation replacement cost	\$75,000,000
Equipment vulnerable to EMP/CME (% of replacement)	55%
Direct damage cost (avoided)	\$41,250,000
Business interruption (6 months at \$2M/day grid revenue)	\$365,000,000
Regulatory penalty for outage	\$5,000,000
Total avoided economic loss	\$411,250,000
Probability-weighted annual avoided loss (2% event probability)	\$8,225,000
CSM value capture rate	10%
CSM system price	\$822,500

Against an EPRI-estimated \$1.2-3.8 million for traditional GMD protection, CSM's \$800K-\$1.2M pricing is competitive while delivering superior protection (dielectric vs. conductive) and lower life-cycle cost (no corrosion maintenance). This is the core pricing narrative: **better protection at lower total cost.**

1.2 Willingness-to-Pay Survey

CSM's pricing analysis incorporates willingness-to-pay (WTP) signals from early customer discussions:

Customer Type	Stated WTP Range	CSM Target Price	% of WTP	Pricing Strategy
DOD Installation	\$2M-\$8M per C4I facility	\$1.5M-\$4M	50-75%	Capture program budget; leave room for procurement margins
Large IOU Substation	\$800K-\$2.5M per substation	\$750K-\$1.5M	60-90%	Competitive with EPRI-estimated alternatives
Commercial Building (\$50M value)	\$200K-\$600K	\$250K-\$450K	75-125%	Premium pricing for early adopters; declines with volume
Maritime Vessel (Container)	\$1.5M-\$4M per vessel	\$1.5M-\$3M	75-100%	Justified by avoided salvage claim (\$10M+)
Luxury Home	\$8K-\$25K	\$10K-\$18K	80-90%	Positioned as home upgrade comparable to solar installation
Dielectric Robot Components	\$15K-\$80K per robot	\$20K-\$60K	75-133%	No existing alternative; price to value

SECTION 2: PRICING BY PRODUCT LINE

2.1 Aegis Grid — Substation Hardening System

Product: Dielectric structural panels, control building envelope, transformer GIC mitigation, and monitoring system for electrical substations.

Pricing Tiers

Tier	Coverage	Substation Size (MVA)	Target Customer	Price
Grid-Essential	Control building only (200-500 sq ft)	<50 MVA	Distribution substations; cooperatives	\$350K-\$550K
Grid-Standard	Control building + transformer GIC mitigation	50-200 MVA	Transmission substations; municipal utilities	\$750K-\$1.25M
Grid-Premium	Full substation envelope + monitoring + 3-year monitoring plan	200-1,000 MVA	Bulk power substations; IOUs	\$1.5M-\$3.0M

Tier	Coverage	Substation Size (MVA)	Target Customer	Price
Grid-Critical	Full protection + redundant monitoring + priority response + annual recertification	>1,000 MVA	Critical NERC-identified substations	\$3.0M-\$6.0M

Volume Discount Schedule

Annual Volume (Substations)	Discount from List
1-2	0%
3-5	8%
6-10	15%
11-25	22%
26+	28%

Recurring Revenue: Annual monitoring and recertification = 8-12% of initial system price per year.

2.2 Aegis Home — Residential Protection System

Product: Whole-home dielectric protection system including main panel dielectric enclosure, smart home electronics protection, and EMP-monitoring sensor with smartphone app.

Pricing by Home Value Tier

Home Value	System Tier	Components	Retail Price	Installed Price (with certified electrician)
\$250K-\$500K	Home-Basic	Main panel protection + 2-room electronics protection	\$4,995	\$6,500-\$7,500
\$500K-\$1M	Home-Standard	Main panel + whole-home electronics + garage EV charger protection	\$8,495	\$10,500-\$12,000
\$1M-\$3M	Home-Premium	Full whole-home dielectric envelope + smart home integration + generator interface	\$14,995	\$18,000-\$22,000
\$3M+	Home-Estate	Custom-engineered full-property protection + satellite building protection + priority monitoring	\$29,995+	\$35,000-\$60,000+

Insurance Premium Offset

Home Tier	Annual Premium Before	Annual Premium After	Annual Savings	Payback Period (Installed)
Home-Basic	\$1,200	\$1,020	\$180	36-42 years (cost-driven; low ROI)
Home-Standard	\$2,500	\$2,125	\$375	28-32 years
Home-Premium	\$7,500	\$6,000	\$1,500	12-15 years
Home-Estate	\$20,000	\$15,000	\$5,000	7-12 years

Home-Basic is priced for accessibility and market-building. Home-Premium and Home-Estate generate attractive standalone ROI. The premium tiers subsidize the basic tier's market development function.

2.3 Aegis Maritime — Vessel Protection System

Product: Dielectric bridge electronics enclosure, navigation system protection, engine control unit protection, and communication system hardening for commercial vessels.

Pricing by Vessel Type

Vessel Type	Vessel Value	Components Protected	System Price
Tugboat / Workboat	\$2M-\$10M	Bridge electronics; engine controls	\$85K-\$175K
Offshore Supply Vessel	\$20M-\$50M	Bridge + dynamic positioning + communications	\$350K-\$650K
Bulk Carrier	\$30M-\$80M	Bridge + engine + navigation + communications	\$500K-\$1.2M
Tanker (Product/Chemical)	\$50M-\$120M	Full bridge + engine + cargo control + communications	\$900K-\$2.2M
Container Ship (>10,000 TEU)	\$150M-\$250M	Full integrated system protection	\$1.5M-\$3.5M
LNG Carrier	\$200M-\$300M	Full protection + cryogenic control system hardening	\$2.5M-\$5.0M
Cruise Ship	\$500M-\$1.5B	Comprehensive protection + passenger safety systems	\$8M-\$20M
Naval Surface Combatant	\$1B-\$5B	Classified protection specification	\$15M-\$50M+

2.4 Aegis Robotics — Dielectric Components

Product: Dielectric actuators, sensors, motor controllers, and communication modules for robotics operating in high-EM/high-radiation environments.

Component	Function	Base Price	Volume Discount (100+ units)
Dielectric Servo Actuator	Position control in high-EM environment	\$8,500	\$5,950
Dielectric Proximity Sensor	Object detection without conductive elements	\$2,200	\$1,540

Component	Function	Base Price	Volume Discount (100+ units)
Dielectric Motor Controller	BLDC motor control for shielded environments	\$12,500	\$8,750
Dielectric Communication Module	RF-isolated data link module	\$4,800	\$3,360
Dielectric Manipulator Arm (5-DOF)	Complete robotic arm for high-EM environments	\$85,000	\$59,500

2.5 Aegis Shield — Premium-Linked Insurance Product

Pricing: CSM does not price the insurance product — insurers price it. CSM receives a platform fee:

Transaction Type	CSM Fee
System installation (policyholder purchase)	CSM sells system at standard pricing
Annual monitoring fee	\$500-\$2,500/year (depending on asset class)
Premium reduction platform fee	5-10% of premium reduction value (paid by insurer as channel commission)
Parametric product origination fee	3-5% of parametric coverage amount
Risk model licensing (insurer SaaS)	\$50K-\$250K/year per carrier, based on portfolio size

SECTION 3: COMPETITIVE PRICING ANALYSIS

3.1 Pricing vs. Faraday Cage Alternatives

Application	Traditional Faraday Solution (Estimated)	CSM Dielectric Solution	Price Premium (CSM)	Life-Cycle Savings (CSM)
Substation control building (500 sq ft)	\$250K-\$500K (copper mesh room)	\$175K-\$350K (dielectric panels)	-30% to -30%	\$50K-\$150K (no corrosion maintenance over 30 years)
C4I facility (2,000 sq ft)	\$6M-\$20M (MIL-SPEC welded steel)	\$1.5M-\$4M (dielectric structural)	-75% to -80%	\$1M-\$3M (maintenance + weight savings in foundation)
Vessel bridge electronics	\$500K-\$2M (copper mesh + surge)	\$200K-\$1.5M (dielectric enclosure + integrated protection)	-60% to -25%	\$50K-\$200K (corrosion + weight)
Home electronics protection	\$2K-\$10K (Faraday bags + surge strips — incomplete)	\$5K-\$30K (whole-home integrated)	+150% to +200%	Vastly superior protection; Faraday bags protect only bagged items

Key Insight: For large-scale applications (C4I facilities, substations, vessels), CSM’s dielectric solution is price-competitive or significantly cheaper than traditional Faraday approaches, even before accounting for life-cycle maintenance savings. The residential price premium is real but justified by the inadequacy of consumer Faraday alternatives.

3.2 Pricing vs. Grid Equipment Solutions

Application	Grid Equipment Solution	CSM Solution	Price Comparison
Transformer GIC blocking	Neutral blocking device: \$150K-\$400K per transformer	Dielectric transformer isolation: \$200K-\$500K	Competitive; dielectric provides additional protection layers
Substation surge protection	Surge arresters + isolation transformers: \$300K-\$800K	Integrated dielectric protection: \$750K-\$1.5M	Higher initial cost; eliminates multiple point-solution vendors
SCADA system protection	Fiber isolation + shielded enclosures: \$100K-\$300K	Dielectric SCADA enclosure: \$80K-\$200K	Competitive to slightly cheaper

SECTION 4: LICENSING AND ALTERNATIVE PRICING MODELS

4.1 Technology Licensing Model

For defense contractors and international partners where direct CSM product sales are impractical:

License Type	Description	Fee Structure
Component License	License to manufacture CSM dielectric panels under CSM patent and quality control	8-12% royalty on licensee's selling price + \$2M-\$10M upfront
Application License	Rights to use CSM dielectric technology in specified application (e.g., military ground vehicles)	technology transfer fee 5-8% royalty + \$5M-\$20M upfront
Geographic License	Exclusive territory rights for non-US markets (e.g., India, Southeast Asia)	\$10M-\$50M upfront + 10-15% royalty
OEM Supply Agreement	CSM manufactures; partner integrates and sells under partner's brand	CSM sells at 35-45% gross margin to partner

4.2 Government Contract Pricing Models

Contract Type	Description	CSM Margin	Risk Profile
Firm Fixed Price (FFP)	Single price for defined deliverable	45-65%	CSM bears performance risk; rewards efficiency
Cost-Plus-Fixed-Fee (CPFF)	Reimbursement of allowable costs + fixed fee (5-15%)	5-15% (fee only)	Low risk; used for R&D and first-of-kind projects
Cost-Plus-Incentive-Fee (CPIF)	Cost reimbursement + incentive fee for meeting/exceeding targets	10-20% (combined)	Moderate risk; incentive for performance
Time & Materials (T&M)	Hourly rates + material costs	25-40% on labor; pass-through on materials	Moderate risk; appropriate for installation services
IDIQ Task Orders	Individual task orders under umbrella contract	Varies by task order type	Mixed; each task order is independent

Recommended Mix: 70% FFP (production contracts), 20% CPFF/CPIF (R&D and development), 10% T&M (installation services).

4.3 Insurance Premium-Linked Pricing

For insurance-distributed products, CSM offers an alternative pricing model where the customer pays nothing upfront:

“Protection-as-a-Service” (PaaS)

Parameter	Utility (Grid) Model	Commercial Property Model	Homeowner Model
Upfront payment	\$0	\$0	\$0
Monthly payment	\$4,000-\$12,000/month	\$1,500-\$4,000/month	\$75-\$250/month
Term	7-10 years	5-10 years	5-10 years
Included services	System, monitoring, maintenance, updates	System, monitoring, annual inspection	System, monitoring, app access
Funded by	Premium savings + operational cost reduction	Premium savings	Premium savings
Residual value	System ownership transfers after term	System ownership transfers after term	System ownership transfers after term
CSM margin (NPV)	50-58% (financing cost of 6-8% built in)	48-55%	45-52%

SECTION 5: PRICE ELASTICITY ESTIMATES

5.1 Price Sensitivity by Segment

Segment	Price Elasticity Estimate	Rationale
DOD Installation Hardening	-0.2 to -0.5 (inelastic)	Mandated by policy; budgets allocated; few alternatives
Utility IOU Substation	-0.5 to -0.8 (moderately inelastic)	Regulatory mandate drives compliance; some cost sensitivity
Utility Cooperative	-0.8 to -1.2 (unit elastic)	Smaller budgets; alternative of self-insurance considered
Commercial Property (via insurance)	-0.3 to -0.6 (inelastic)	Insurance requirement creates non-discretionary demand
Maritime Vessel	-0.6 to -1.0 (moderately elastic)	Classification society requirement drives compliance; retrofit timing discretionary
Home Builder	-0.8 to -1.5 (elastic)	Optional upgrade; price-sensitive consumer market
Homeowner (Direct)	-1.0 to -2.0 (highly elastic)	Discretionary purchase; premium savings soften elasticity
Dielectric Robot Components	-0.3 to -0.7 (inelastic)	No substitute products exist for dielectric functionality

5.2 Revenue Optimization

Given these elasticities, CSM’s revenue-optimal pricing strategy varies by segment:

- **Inelastic segments (Government, Utility, Insurance-linked):** Price at 75-100% of WTP; value capture rate of 10-15%. These segments are revenue anchors.
- **Moderately elastic segments (Maritime, Robot Components):** Price at 60-80% of WTP; value capture of 8-12%. Volume growth potential.

- **Elastic segments (Home Builder, Homeowner):** Price at 40-60% of WTP; value capture of 5-8%. Market share growth prioritized over per-unit margin; insurance premium savings create effective price reduction.

SECTION 6: GEOGRAPHIC PRICING VARIATIONS

6.1 Regional Price Multipliers

Region	Price Multiplier (vs. US Base)	Rationale
United States (Base)	1.00x	Primary market; regulatory tailwinds; reference pricing
Canada	0.90x	Currency adjustment; familiarity with US pricing
United Kingdom	1.15x	Higher construction costs; strong regulatory environment
European Union (core)	1.10x	CER Directive compliance; higher labor costs for installation
Japan	1.25x	High labor costs; premium for reliability in earthquake-prone grid
South Korea	1.05x	Competitive manufacturing environment; strong semiconductor protection demand
Australia/New Zealand	1.10x	Distance/shipping costs; CME exposure in southern hemisphere
GCC States (UAE, Saudi Arabia)	1.30x	Premium for resilience in critical infrastructure; low price sensitivity
India	0.55x	Price-sensitive market; licensing model preferred over direct sales
Southeast Asia	0.65x	Developing infrastructure market; price sensitivity
Latin America	0.70x	Mixed market; Brazil/Chile premium segments; others price-sensitive

6.2 Currency and Inflation Risk

CSM will price international contracts in USD where possible, with periodic price adjustment mechanisms for contracts denominated in other currencies. Inflation adjustment clauses (CPI-U + 2% or similar) will be included in multi-year contracts to protect margins.

SECTION 7: GROSS MARGIN ANALYSIS

7.1 Product-Level Gross Margins (At Scale)

Product	Materials	Labor/Certification	Warranty Reserve	COGS Total	Price	Gross Margin
Grid-Standard (per substation)	\$180K-\$300K	\$100K-\$180K	\$30K-\$50K	\$310K-\$530K	\$750K-\$1.25M	55-60%
Home-Premium (installed)	\$4.5K-\$6K	\$2K-\$3.5K	\$500-\$800	\$7K-\$10.3K	\$18K-\$22K	50-55%

Product	Materials	Labor/Certification	Warranty Reserve	COGS Total	Price	Gross Margin
Maritime-Container Ship	\$450K-\$900K	\$250K-\$500K	\$75K-\$150K	\$775K-\$1.55M	\$1.5M-\$3.5M	48-56%
Robot Actuator (100+ units)	\$1.2K-\$1.8K	\$800-\$1.2K	\$200-\$300	\$2.2K-\$3.3K	\$5.95K	45-55%
Aegis Shield Platform Fee	Near-zero marginal cost	—	—	<5% of fee	Variable	95%+

Weighted Average Gross Margin (5-Year Projection): 55-62%

7.2 Margin Improvement Trajectory

Factor	Year 1	Year 3	Year 5	Year 10
Material costs (volume purchasing)	Base	-8%	-15%	-22%
Labor efficiency (learning curve)	Base	-12%	-25%	-35%
Automation investment	Base	Base	-10%	-25%
Warranty cost (reliability data)	8% reserve	6% reserve	4% reserve	3% reserve
Combined GM Improvement	38%	48%	58%	65%

SECTION 8: PRICING RISK FACTORS

1. **Price transparency risk:** As CSM products become specified in government and utility RFPs, pricing may need to be disclosed (FAR requirement for FFP contracts), potentially creating pricing pressure across segments.
2. **Commoditization risk:** If dielectric panels become commodity items (multiple manufacturers, interchangeable specifications), margins compress toward manufacturing cost. Mitigation: CSM's systems integration, monitoring, and certification layers capture value beyond the material.
3. **Insurance premium fluctuation:** If insurance premiums for hardened assets don't decline as modeled, the PaaS model's economics break. CSM's risk is limited to PaaS contracts; direct-sale customers bear premium fluctuation risk.
4. **Regulatory price pressure:** If FERC or state PUCs impose cost caps on EMP hardening (analogous to rate caps on medical devices), margins could compress. CSM's value-based pricing narrative argues against price controls — the product prevents losses that dwarf its cost.
5. **Currency risk on international contracts:** Long-duration international contracts denominated in local currency carry exchange rate risk. CSM will attempt to denominate in USD and include FX adjustment clauses.

Document prepared by: Carrington Storm Motors Finance & Strategy Data sources: EPRI, NERC, DOD MILCON cost data, RSMeans construction cost data, insurance industry premium surveys, marine industry newbuild cost data Last updated: July 15, 2026